

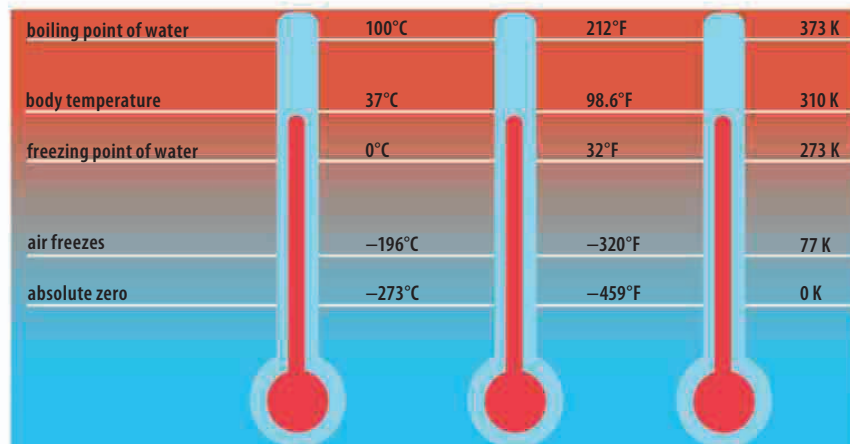
Heat transfer

THERMODYNAMICS IS THE STUDY OF THE WAY HEAT MOVES FROM ONE SUBSTANCE TO ANOTHER.

Heat is the name used for the type of energy that makes the atoms and molecules move inside a substance. Adding energy makes these particles move more quickly—and results in the substance heating up.

Measuring heat

Temperature is a measure of the heat in a substance. It is an average figure for the energy contained by every particle. Temperature and energy are not interchangeable. A spark from a fire can have a very high temperature, but it does not cause much of a burn because it contains only a small amount of energy. Temperature is measured using a scale. The difference between the upper and lower points is divided into a fixed number of units, or degrees, and any temperature can be expressed in multiples of degrees.



△ What happens?

This scale shows what happens at some significant temperatures.

△ Celsius

Water freezes at 0°C and boils at 100°C. Celsius was previously known as centigrade.

△ Fahrenheit

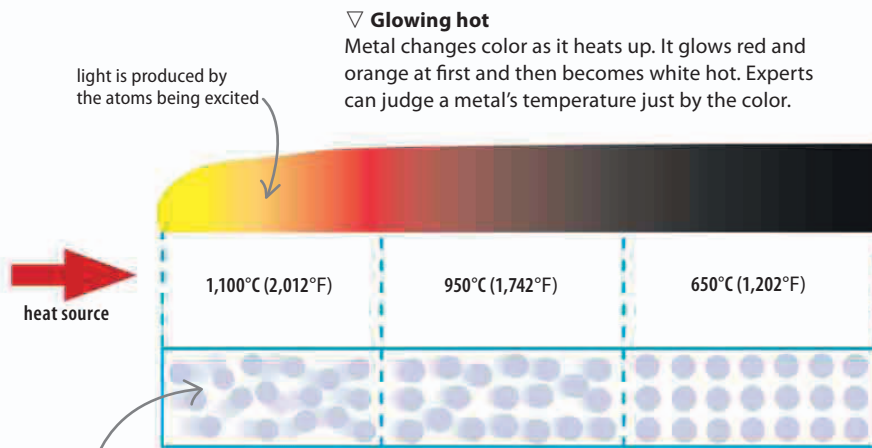
The Fahrenheit scale starts from the freezing point of saturated saltwater (0°F).

△ Kelvin

Absolute zero (0 K) is the temperature at which all particles cease to move completely.

Conduction

Heat always moves from areas with high thermal energy to areas with less. In other words, hot things always cool down, while cold things warm up to match their surroundings. Heat moves through a solid by conduction. This is a phenomenon in which the motion of particles in the hot part of the solid gradually transfers to neighboring ones, making them move faster and sending heat energy through the solid. Metals conduct heat better than nonmetals because their electrons are more free to move and pass their energy on.



▽ Glowing hot

Metal changes color as it heats up. It glows red and orange at first and then becomes white hot. Experts can judge a metal's temperature just by the color.

△ Hot metal

The vibrational motion of the metal atoms is proportional to the heat energy they contain. Those in the warmest part of the bar move faster than those in the cooler areas.